

Retained Loop Projection and Branch-History Monodromy in Transdimensional Anomalous Hall Transport

Justin K. Lietz
Neuroca, Inc.

May 1, 2026

Abstract

The transdimensional anomalous Hall effect (TDAHE) in rhombohedral thin graphite reports an anomalous Hall state that couples both out-of-plane and in-plane orbital magnetizations. The experimental finding is structurally unusual because the in-plane orbital magnetization is generated by a finite-thickness orbital loop and leaves a Hall hysteresis witness that the ordinary two-dimensional anomalous Hall projector does not retain. This paper develops a full Phase Calculus reading of that phenomenon. The argument is not a microscopic derivation of the graphene Hamiltonian, the Hartree–Fock crescent Fermi surface, or the measured material constants. It is a retained-state explanation of why the anomaly has the form it has, why the ordinary two-dimensional Hall quotient is incomplete, and which new experiments should distinguish a genuine branch-history mechanism from a scalar hysteresis model. The paper builds the required Phase Calculus machinery from first principles: primitive roll, lifted state, the operator core Q, B, L , CF000 saturation and forced orthogonal re-articulation, quotient descent $\Pi \circ E = G \circ \Pi$, and Monodromy-as-Memory. It then proves that a through-thickness loop can have identical planar projection while carrying opposite in-plane orbital moment; that an ordinary $2D$ Hall projector cannot recover the transdimensional residual; and that two field-training operations acting on independent hysteresis registers generically produce a nonzero field-order commutator. The resulting extension program proposes direct measurements: quotient residual extraction, field-order commutators, angular tomography of the loop branch, coherence-collapse scaling, symmetry-restoration suppression tests, and dual-memory minor-loop surfaces.

Keywords: Phase Calculus, transdimensional anomalous Hall effect, anomalous Hall effect, monodromy, hysteresis, branch memory, projection loss, orbital magnetization, rhombohedral graphene, quotient descent

Contents

1	Thesis and scope	2
1.1	The claim	2
1.2	What this paper does not claim	3
1.3	Why Phase Calculus must be built before the mapping	3
2	Experimental target: what must be structurally explained	4
2.1	Ordinary AHE projector	4
2.2	The transdimensional regime	4
2.3	The exact burden	4
3	Phase Calculus I: primitive roll and visible insufficiency	5
3.1	Primitive roll	5
4	Phase Calculus II: the lifted state and the operator core	5
4.1	Lifted state	5
4.2	Operator core	6
4.3	Exact quotient criterion	6

5	Phase Calculus III: CF000 saturation	7
5.1	Same-axis saturation	7
5.2	TDAHE as saturation	7
6	Retained loop theorem for the in-plane orbital moment	7
7	Ordinary Hall quotient nonclosure and transdimensional residual	8
7.1	Two projectors	8
8	Monodromy-as-Memory: branch history as algebra	9
8.1	Abstract monodromy register	9
9	TDAHE hysteresis as monodromy memory	11
9.1	Field protocols as generators	11
9.2	Field-order commutator	11
9.3	Projection obstruction for scalar memory	12
10	Exact structural crosswalk	12
11	Figure set: retained-branch structural replication	13
12	Experimental consequences of retained-branch transport	17
12.1	Vector switching surface	17
12.2	Field-order memory	17
12.3	Angular tomography of the retained loop branch	17
12.4	Coherence-collapse scaling	18
12.5	Dual-memory minor-loop surface	18
12.6	First measurement package	18
12.7	Distinguishing outcomes	19
13	Verification and reproducibility support	19
14	Chronology and provenance	19
15	Discussion	20
15.1	Why the match is stronger than analogy	20
15.2	Why the ordinary AHE rule is not state-complete	20
15.3	Why monodromy is the right memory language	20
15.4	Relation to the experimental microscopic picture	20
16	Conclusion	20
A	Appendix A: compact theorem list	21
B	Appendix B: symbolic verification targets	21
C	Appendix C: source ledger template	22

1 Thesis and scope

1.1 The claim

This paper claims that the transdimensional anomalous Hall effect is a material witness of a specific Phase Calculus pattern:

same-axis saturation $\longrightarrow$ non-discharge $\longrightarrow$ orthogonal re-articulation $\longrightarrow$ retained branch residual.

The TDAHE paper reports an anomalous Hall regime in rhombohedral thin graphite in which carriers sustain coherent orbital motion both within and across the material plane, producing anomalous Hall response coupled to both M_z and M_y [1, 2]. The ordinary anomalous Hall grammar expects an out-of-plane orbital magnetization associated with in-plane chiral orbital motion. The transdimensional regime admits an additional branch: a through-thickness orbital loop whose moment lies in plane. In the experimental geometry this produces pronounced Hall resistance hysteresis controlled by both out-of-plane and in-plane magnetic fields [1, 2].

Phase Calculus does not replace the microscopic condensed-matter analysis. It supplies a state-complete language for the part of the phenomenon that is otherwise described as an anomaly: the ordinary projector has discarded a branch that the material state still carries.

The compressed statement is:

TDAHE is a saturation-driven, loop-generated, branch-history-bearing projection residual.

Each phrase is load-bearing.

1. **Saturation-driven:** CF000 states that when same-axis articulation is exhausted and discharge is excluded, continuation is forced into an orthogonal articulation class.
2. **Loop-generated:** the new branch is not a static hidden parameter; it is produced by a retained orbital loop.
3. **Branch-history-bearing:** the two magnetic training directions write hysteresis memory into different retained registers.
4. **Projection residual:** ordinary two-dimensional AHE projection erases the in-plane loop branch; the residual is the measurable difference between the retained transdimensional witness and the stripped $2D$ witness.

1.2 What this paper does not claim

This paper does not claim:

1. that Phase Calculus derives the rhombohedral graphene Hamiltonian;
2. that Phase Calculus derives the Hartree–Fock crescent Fermi surface;
3. that Phase Calculus predicts the material values of carrier density, displacement field, coercive field, or conductivity without material input;
4. that the quoted experimental phenomenon proves the entire Phase Calculus program;
5. that every Hall anomaly is automatically a Phase Calculus branch residual.

The claim is narrower and stronger: once the TDAHE material state is known to retain a coherent through-thickness orbital loop, Phase Calculus gives a theorem-grade account of why ordinary two-dimensional projection is not state-complete and why branch-history experiments should reveal noncommuting field-order memory.

1.3 Why Phase Calculus must be built before the mapping

A reader cannot evaluate the mapping if Phase Calculus appears only as vocabulary. The required objects are not names. They are a chain of commitments:

$$\text{primitive roll} \rightarrow \widehat{\Xi} \rightarrow \{Q, B, L\} \rightarrow \text{saturation} \rightarrow \Pi \circ E = G \circ \Pi \rightarrow \text{branch-history register}.$$

The paper therefore begins by constructing these objects before applying them to the material system.

2 Experimental target: what must be structurally explained

2.1 Ordinary AHE projector

In the ordinary two-dimensional anomalous Hall picture, the relevant orbital loop lies in the material plane. Its orbital magnetization is out of plane. If current flows along x , the conventional Hall witness is organized by an out-of-plane magnetization M_z . At the level of a reduced state, one can write:

$$X_{2D} = (M_z, h_z),$$

where h_z is the out-of-plane hysteresis branch memory. A minimal projected Hall witness is

$$R_{2D}(X_{2D}) = a_z h_z.$$

This is not a microscopic formula. It is the reduced memory form needed for the proof: the ordinary projector retains the branch that writes out-of-plane hysteresis and discards any branch not expressible through that state.

2.2 The transdimensional regime

The TDAHE experiment identifies an intermediate thickness regime. The sample thickness is much larger than atomic spacing but comparable to a vertical coherence-transport or mean-free length. In this regime, the carriers can sustain coherent orbital motion both within and across the plane [1, 2]. The paper reports a state coupled to both in-plane and out-of-plane orbital magnetizations and exhibiting Hall resistance hysteresis under both in-plane and out-of-plane magnetic fields [1].

The reduced retained state therefore requires at least

$$X_{TD} = (M_z, M_y, h_z, h_y, \chi),$$

where:

- M_z is the conventional out-of-plane orbital magnetization branch;
- M_y is the in-plane orbital magnetization branch;
- h_z is the memory written by the out-of-plane field protocol;
- h_y is the memory written by the in-plane field protocol;
- $\chi \in \{0, 1\}$ is the branch-activity selector for the transdimensional loop sector.

A minimal transdimensional witness is

$$R_{TD}(X_{TD}) = a_z h_z + \chi a_y h_y.$$

If $\chi a_y h_y \neq 0$, then the ordinary $2D$ quotient has lost a physically active branch.

2.3 The exact burden

The structural problem is not simply that TDAHE has another term. The problem is that the new term is generated by a loop whose projection into the old plane can be erased while its orbital moment remains nonzero. Therefore the correct mathematical target is:

same planar projection $\nrightarrow$ same retained loop state.

This is the exact setting for Phase Calculus: visible recurrence or visible coincidence does not imply full retained-state identity.

3 Phase Calculus I: primitive roll and visible insufficiency

3.1 Primitive roll

Phase Calculus starts from a retained roll, not from a scalar point. The primitive visible roll is

$$z(0) = i, \quad \frac{dz}{d\theta} = iz, \quad z(\theta) = ie^{i\theta}.$$

The visible quarter witnesses are

$$z(0) = i, \quad z(\pi/2) = -1, \quad z(\pi) = -i, \quad z(3\pi/2) = 1, \quad z(2\pi) = i.$$

The visible carrier returns after 2π , but the visible carrier is not the full state. The same visible value can occur with different completed-turn memory, branch state, or germ width. The distinction between visible recurrence and retained identity is the first nonnegotiable premise.

Definition 3.1 (Visible witness). Let a lifted state carry phase θ . Its visible witness is

$$V(\widehat{\Xi}) = ie^{i\theta}.$$

Definition 3.2 (Completed-turn memory). The completed-turn memory is

$$\kappa = \left\lfloor \frac{\theta}{2\pi} \right\rfloor.$$

Two states can have the same visible witness and different κ .

Proposition 3.1 (Visible recurrence is not retained identity). *There exist distinct retained states $\widehat{\Xi}_a \neq \widehat{\Xi}_b$ with $V(\widehat{\Xi}_a) = V(\widehat{\Xi}_b)$.*

Proof. Let $\theta_b = \theta_a + 2\pi m$ for some nonzero integer m . Then

$$ie^{i\theta_b} = ie^{i\theta_a} e^{i2\pi m} = ie^{i\theta_a}.$$

But

$$\left\lfloor \frac{\theta_b}{2\pi} \right\rfloor = \left\lfloor \frac{\theta_a}{2\pi} \right\rfloor + m$$

whenever θ_a is not on a floor-boundary ambiguity; in the standard retained state the completed-turn coordinate differs. Thus the visible witness agrees while retained memory differs. $\square$

This theorem is the formal ancestor of the TDAHE projection problem. Ordinary Hall readout can agree on the branch it retains while not determining a branch it discards.

4 Phase Calculus II: the lifted state and the operator core

4.1 Lifted state

The baseline Phase Calculus lifted state is

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right].$$

The coordinates play distinct roles:

- A : host/articulation class;
- $q = (u, v)$: balanced denominator-pair resolution state;
- θ : phase coordinate;

- κ : completed-turn memory;
- c : completion germ or local retained uncertainty interval.

The point is not that every physical application literally uses the same five coordinates. The point is that Phase Calculus demands a retained state whose projection is a boundary readout, not the whole system.

4.2 Operator core

The primitive operator alphabet is

$$\mathcal{U} = \{Q, B, L\}.$$

The three operators are:

$$Q(A, q, \theta, \kappa, c) = \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right), \quad (1)$$

$$B(A, (u, v), \theta, \kappa, c) = \left(A, \text{sort}(v, u + v), \theta, \kappa, C\left(\theta, \frac{1}{v(u + v)}\right) \right), \quad (2)$$

$$L(A, q, \theta, \kappa, c) = \left(N(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right), \quad (3)$$

where

$$C(\theta, r) = [\theta - \pi r, \theta + \pi r].$$

The meanings are:

- Q : same-host quarter continuation;
- B : balanced refinement of the retained resolution corridor;
- L : host lift or orthogonal re-articulation.

The TDAHE application uses the L -type idea structurally: a saturated lower-dimensional articulation admits an orthogonal branch. It also uses the Q -type idea structurally: the branch is loop-generated and phase/history bearing. It uses the quotient theorem to show why one projector is incomplete while the retained projector closes.

4.3 Exact quotient criterion

Definition 4.1 (Exact quotient). Let $E : X \rightarrow X$ be retained evolution and let $\Pi : X \rightarrow Y$ be a projector. A visible branch law $G : Y \rightarrow Y$ is an exact quotient of E through Π if

$$\Pi \circ E = G \circ \Pi.$$

Definition 4.2 (Residual). For a candidate visible law G , define

$$\mathcal{R}_{\Pi, G}(X) = \Pi(EX) - G(\Pi X).$$

Proposition 4.1 (Exactness criterion). *The visible law G is exact on the retained regime if and only if $\mathcal{R}_{\Pi, G}(X) = 0$ for all retained states X in the regime.*

Proof. This is immediate from the definition:

$$\mathcal{R}_{\Pi, G}(X) = 0 \iff \Pi(EX) = G(\Pi X)$$

for all X , which is exactly $\Pi \circ E = G \circ \Pi$. $\square$

The ordinary 2D Hall quotient does not close on the TDAHE branch because its residual is the in-plane branch contribution. The transdimensional quotient restores closure by retaining the missing branch.

5 Phase Calculus III: CF000 saturation

5.1 Same-axis saturation

CF000 supplies the saturation theorem used in this paper. It can be expressed without importing any material coordinate system.

Definition 5.1 (Same-axis saturation). An admitted axis or articulation class is same-axis saturated when no genuinely new lawful invariant-bearing articulation remains available within that same class.

Assumption 5.1 (Non-discharge). The carried invariant may not terminate into a forbidden terminal state merely because the current articulation class is saturated.

Theorem 5.1 (CF000 saturation theorem). *Once lawful same-axis expression is exhaustively saturated, further admitted articulation in that same class is impossible without contradiction. If terminal discharge is excluded, orthogonal re-articulation is forced.*

This theorem is not a claim about Euclidean perpendicularity. It is a claim about articulation classes. A new axis is “orthogonal” when it is irreducible to the saturated class but still bears the continuing invariant.

5.2 TDAHE as saturation

The ordinary 2D AHE quotient retains the in-plane orbital loop branch that produces M_z . It is saturated relative to the TDAHE evidence because it cannot express a Hall hysteresis branch controlled by $B_{\parallel}$ and generated by in-plane orbital magnetization M_y . The Hall response does not disappear. A new branch appears: coherent through-thickness orbital motion generating an in-plane orbital moment.

The structural map is:

CF000 structure	TDAHE realization
Same-axis saturation	Ordinary 2D AHE projection retains only the in-plane chiral loop branch producing M_z .
Non-discharge	The anomalous Hall response persists and exhibits robust hysteresis.
Orthogonal re-articulation	A through-thickness orbital loop is retained, producing M_y .
New branch witness	$B_{\parallel}$ -controlled Hall hysteresis appears.
Projection residual	$R_{TD} - R_{2D} = \chi a_y h_y$.

The claim is not that CF000 derives the material mechanism. The claim is that the material mechanism instantiates the CF000 structural pattern.

6 Retained loop theorem for the in-plane orbital moment

The paper now proves the minimal mathematical fact behind the retained-loop reading. A loop can have a planar projection that loses its through-thickness orientation while preserving a nonzero in-plane orbital moment in the retained state.

Definition 6.1 (Retained through-thickness loop). For $a, b > 0$ and orientation $\varepsilon \in \{-1, +1\}$, define a loop in the xz -plane by

$$\gamma_{\varepsilon}(t) = (a \cos t, 0, \varepsilon b \sin t), \quad t \in [0, 2\pi].$$

Definition 6.2 (Planar material projection). Let $P_0 : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the material-plane projection

$$P_0(x, y, z) = (x, y).$$

Definition 6.3 (Orbital magnetic moment of a current loop). For a current I carried by a closed loop γ , define its magnetic moment by

$$\boldsymbol{\mu}(\gamma) = \frac{I}{2} \oint_{\gamma} \mathbf{r} \times d\mathbf{r}.$$

Theorem 6.1 (Planar projection erases the orientation of the through-thickness loop). *The two loops γ_+ and γ_- have identical material-plane projection:*

$$P_0\gamma_+(t) = P_0\gamma_-(t)$$

for all t . Their in-plane orbital moments are opposite:

$$\boldsymbol{\mu}(\gamma_{\varepsilon}) = -\varepsilon\pi Iab \mathbf{e}_y.$$

Proof. The projection is immediate:

$$P_0\gamma_{\varepsilon}(t) = (a \cos t, 0),$$

which is independent of ε .

For the moment, compute

$$\mathbf{r}(t) = (a \cos t, 0, \varepsilon b \sin t), \quad \mathbf{r}'(t) = (-a \sin t, 0, \varepsilon b \cos t).$$

Then

$$\mathbf{r} \times \mathbf{r}' = \begin{pmatrix} 0 \\ zx' - xz' \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ (\varepsilon b \sin t)(-a \sin t) - (a \cos t)(\varepsilon b \cos t) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -\varepsilon ab \\ 0 \end{pmatrix}.$$

Therefore

$$\boldsymbol{\mu} = \frac{I}{2} \int_0^{2\pi} (0, -\varepsilon ab, 0) dt = (0, -\varepsilon\pi Iab, 0) = -\varepsilon\pi Iab \mathbf{e}_y.$$

□

Corollary 6.2 (No planar loop quotient is state-complete for the retained moment). *No projector that sees only $P_0\gamma$ can determine $\mu_y(\gamma)$ on the retained loop family $\{\gamma_+, \gamma_-\}$.*

Proof. Both loops have the same $P_0\gamma$, but they have opposite μ_y . A function of $P_0\gamma$ alone must assign the same value to both loops, while μ_y assigns different values. □

This theorem is the clean mathematical core. TDAHE is not merely “another magnetization.” It is a retained loop branch whose planar projection can erase the very orientation that generates M_y .

7 Ordinary Hall quotient nonclosure and transdimensional residual

7.1 Two projectors

Define the ordinary 2D projector:

$$\Pi_{2D}(M_z, M_y, h_z, h_y, \chi) = (M_z, h_z).$$

Define the transdimensional projector:

$$\Pi_{TD}(M_z, M_y, h_z, h_y, \chi) = (M_z, M_y, h_z, h_y, \chi).$$

The ordinary Hall witness is

$$R_{2D} = a_z h_z.$$

The transdimensional Hall witness is

$$R_{\text{TD}} = a_z h_z + \chi a_y h_y.$$

The residual is

$$\Delta_{\text{TD}} = R_{\text{TD}} - R_{2D} = \chi a_y h_y.$$

If one explicitly includes the loop moment, write

$$R_{\text{TD}} = a_z h_z + \chi a_y h_y \mu_y(\gamma),$$

so that

$$\Delta_{\text{TD}} = \chi a_y h_y \mu_y(\gamma).$$

Theorem 7.1 (Ordinary AHE projection is not state-complete on the TDAHE branch). *If $\chi a_y h_y \neq 0$, then the ordinary 2D AHE projector cannot determine the full transdimensional Hall witness.*

Proof. Let

$$X = (M_z, M_y, h_z, h_y, \chi).$$

Then

$$R_{2D}(\Pi_{2D} X) = a_z h_z,$$

while

$$R_{\text{TD}}(X) = a_z h_z + \chi a_y h_y.$$

If $\chi a_y h_y \neq 0$, these differ. Therefore the ordinary projected state does not determine the full witness. $\square$

Theorem 7.2 (Exact residual theorem). *For the reduced TDAHE retained state, the nonclosure of the ordinary 2D quotient is exactly the retained in-plane branch term:*

$$\Delta_{\text{TD}}(X) = \chi a_y h_y.$$

With explicit loop moment,

$$\Delta_{\text{TD}}(X, \gamma) = \chi a_y h_y \mu_y(\gamma).$$

Proof. Subtract the ordinary witness from the retained witness:

$$R_{\text{TD}} - R_{2D} = (a_z h_z + \chi a_y h_y) - a_z h_z = \chi a_y h_y.$$

The loop-moment form is the same computation after replacing the in-plane branch amplitude by its loop-generated moment factor. $\square$

This theorem is the algebraic version of the experimental surprise. Nothing is anomalous in the retained state. The anomaly is the residual produced by reading a transdimensional state through a lower-dimensional quotient.

8 Monodromy-as-Memory: branch history as algebra

8.1 Abstract monodromy register

The TDAHE hysteresis problem is not fully expressed by static coordinates (M_z, M_y) . Hysteresis means path memory. Phase Calculus carries path memory using a branch-history register.

Definition 8.1 (Branch cover and monodromy action). Let S be a sheet set and G a group of loop or protocol classes. A monodromy action is a homomorphism

$$\rho : G \rightarrow \text{Sym}(S).$$

Definition 8.2 (Branch-complete lifted state). A branch-complete lifted state is

$$\widehat{\Xi}_P = (A, q, \theta, \kappa, c, s, h),$$

where $s \in S$ is the active sheet and $h \in G$ is the retained branch-history word.

Definition 8.3 (Loop transport). For $g \in G$, define

$$E_g(A, q, \theta, \kappa, c, s, h) = (A, q, \theta, \kappa + \omega(g), c, \rho(g)(s), hg),$$

where $\omega(g)$ is any completed-turn contribution assigned to the loop class.

Theorem 8.1 (Composition law). *Under the convention that gg' means “first traverse g , then traverse g' ,”*

$$E_{gg'} = E_{g'} \circ E_g.$$

Proof. Apply E_g first:

$$(A, q, \theta, \kappa, c, s, h) \mapsto (A, q, \theta, \kappa + \omega(g), c, \rho(g)s, hg).$$

Apply $E_{g'}$:

$$(A, q, \theta, \kappa + \omega(g) + \omega(g'), c, \rho(g')\rho(g)s, hgg').$$

By the group-action convention and additivity of the completed-turn contribution on the retained loop classes, this equals $E_{gg'}$. $\square$

Theorem 8.2 (Sheet quotient commutation). *Let $\Pi_{\text{sheet}}(\widehat{\Xi}_P) = s$. Then*

$$\Pi_{\text{sheet}} \circ E_g = \rho(g) \circ \Pi_{\text{sheet}}.$$

Proof. By definition,

$$\Pi_{\text{sheet}}(E_g(A, q, \theta, \kappa, c, s, h)) = \Pi_{\text{sheet}}(A, q, \theta, \kappa + \omega(g), c, \rho(g)s, hg) = \rho(g)s.$$

This is $\rho(g)(\Pi_{\text{sheet}}\widehat{\Xi}_P)$. $\square$

Theorem 8.3 (Full-register faithfulness). *If $E_g = E_{g'}$ as full-register maps on all lifted states, then $g = g'$ in G . Conversely, if $g = g'$, then $E_g = E_{g'}$.*

Proof. The converse is immediate. For the forward direction, take any lifted state with branch history $h = e$, the identity. The history coordinate after applying E_g is g ; the history coordinate after applying $E_{g'}$ is g' . If the full-register maps are equal for all states, these output histories are equal. Hence $g = g'$. $\square$

Corollary 8.4 (Stripped projection is not faithful). *Any projection that discards h cannot in general distinguish two different loop words $g \neq g'$ when their visible sheet action or visible scalar witness happens to agree.*

Proof. The full-register theorem distinguishes g and g' by the history coordinate. A stripped projection removes that coordinate. Therefore it cannot be faithful to the full register in general. $\square$

This is the algebraic spine for the TDAHE field-order experiment. Field sweeps are not merely values of an external field. They are operations that write words into the retained hysteresis register.

9 TDAHE hysteresis as monodromy memory

9.1 Field protocols as generators

Let g_z denote an out-of-plane field-training operation, and let g_y denote an in-plane field-training operation. In a single scalar hysteresis model, the two protocols are forced into one memory line. In a retained TDAHE model, they act on two registers:

$$h = \begin{pmatrix} h_z \\ h_y \end{pmatrix}.$$

A minimal coupled linear update model is

$$U_z = \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix}, \quad U_y = \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}.$$

Here:

- U_z writes the z -branch and can drag the y -branch through coupling α ;
- U_y writes the y -branch and can drag the z -branch through coupling β ;
- α, β are coupling coefficients, not assumed equal.

The Hall readout is

$$R(h) = a_z h_z + a_y h_y.$$

The sign convention follows the reduced witness form used earlier in the paper. Reversing the orientation convention for the y -branch changes the sign of the $a_y h_y$ term, not the existence of the order-dependent commutator.

9.2 Field-order commutator

Theorem 9.1 (Nonzero field-order commutator). *For*

$$U_z = \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix}, \quad U_y = \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix},$$

the matrix commutator difference is

$$U_y U_z - U_z U_y = \alpha\beta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Proof. Compute:

$$U_y U_z = \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} = \begin{pmatrix} 1 + \alpha\beta & \beta \\ \alpha & 1 \end{pmatrix}.$$

Also,

$$U_z U_y = \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \beta \\ \alpha & 1 + \alpha\beta \end{pmatrix}.$$

Subtracting gives

$$U_y U_z - U_z U_y = \begin{pmatrix} \alpha\beta & 0 \\ 0 & -\alpha\beta \end{pmatrix}.$$

□

Theorem 9.2 (Measurable Hall commutator). *Let $R(h) = a_z h_z + a_y h_y$. Define*

$$\mathcal{C}_{zy}(h) = R(U_y U_z h) - R(U_z U_y h).$$

Then

$$\mathcal{C}_{zy}(h) = \alpha\beta(a_z h_z - a_y h_y).$$

Proof. From the previous theorem,

$$(U_y U_z - U_z U_y)h = \begin{pmatrix} \alpha\beta & 0 \\ 0 & -\alpha\beta \end{pmatrix} \begin{pmatrix} h_z \\ h_y \end{pmatrix} = \begin{pmatrix} \alpha\beta h_z \\ -\alpha\beta h_y \end{pmatrix}.$$

Applying $R(h) = a_z h_z + a_y h_y$ to the difference gives

$$\mathcal{C}_{zy}(h) = a_z(\alpha\beta h_z) + a_y(-\alpha\beta h_y) = \alpha\beta(a_z h_z - a_y h_y).$$

□

Remark 9.1. If the y -branch Hall sign is encoded with the opposite orientation, the same derivation changes the sign of the $a_y h_y$ contribution. The experimentally meaningful claim is not the convention. The claim is the nonzero order-dependent residual proportional to $\alpha\beta$.

9.3 Projection obstruction for scalar memory

Theorem 9.3 (No scalar h_z -only closure under U_y when $\beta \neq 0$). *Let $p_z(h_z, h_y) = h_z$. If $\beta \neq 0$, there is no map G_y on the scalar h_z -coordinate alone such that*

$$p_z \circ U_y = G_y \circ p_z$$

for all $h = (h_z, h_y)$.

Proof. Compute

$$U_y(h_z, h_y) = (h_z + \beta h_y, h_y).$$

Therefore

$$p_z(U_y(h_z, h_y)) = h_z + \beta h_y.$$

If a scalar map G_y existed with $p_z \circ U_y = G_y \circ p_z$, then $G_y(h_z)$ would have to equal $h_z + \beta h_y$ for all h_y . This is impossible when $\beta \neq 0$, because the right-hand side depends on h_y while the left-hand side does not. □

This theorem is the clean algebraic reason the ordinary Hall projection cannot close the TDAHE field-training dynamics. The missing branch is not optional. Once $B_{||}$ affects the retained state, a scalar h_z -only memory law cannot reproduce the protocol.

10 Exact structural crosswalk

The following table records the full correspondence used in this paper.

Phase Calculus object	TDAHE object
Primitive roll $z(\theta) = ie^{i\theta}$	Retained orbital loop whose projected transport produces a visible Hall witness.
Visible witness not state-complete	Ordinary Hall projection reads a lower-dimensional witness and loses the in-plane loop branch.
Lifted state $(A, q, \theta, \kappa, c)$	Retained material state containing host, loop class, branch memories, and coherence selector.
Operator Q	Loop/quarter continuation: same-host cyclic transport.
Operator B	Refinement or resolution of branch/witness under finite resources; not central to microscopic graphene, but relevant to retained-state tightening.
Operator L	Orthogonal re-articulation: admission of a new branch when same-axis expression saturates.
CF000 saturation theorem	Ordinary 2D AHE quotient cannot express $B_{\parallel}$ -controlled Hall hysteresis; an orthogonal through-thickness loop branch is admitted.
Quotient law $\Pi \circ E = G \circ \Pi$	Π_{TD} closes the retained transport; Π_{2D} leaves a residual.
Monodromy history $h \in G$	Hysteresis memory word written by $B_{\perp}$ and $B_{\parallel}$ field protocols.
Noncommuting loop words	Field-order commutator $\mathcal{C}_{zy} \neq 0$.
Stripped projection nonfaithfulness	Ordinary Hall readout can miss the order information carried by the two-register branch state.

This is not merely a list of analogies. Each row is tied to either a retained-state identity, a quotient equation, or a proposed measurement.

11 Figure set: retained-branch structural replication

The figures in this section are model-output figures, not digitized experimental traces. Their purpose is to make the retained-branch mechanism visually concrete: the full transdimensional state keeps a loop branch that the ordinary two-dimensional quotient discards. The companion materials provide deterministic tables, symbolic checks, and figure-generation notebooks for reproducing the panels.

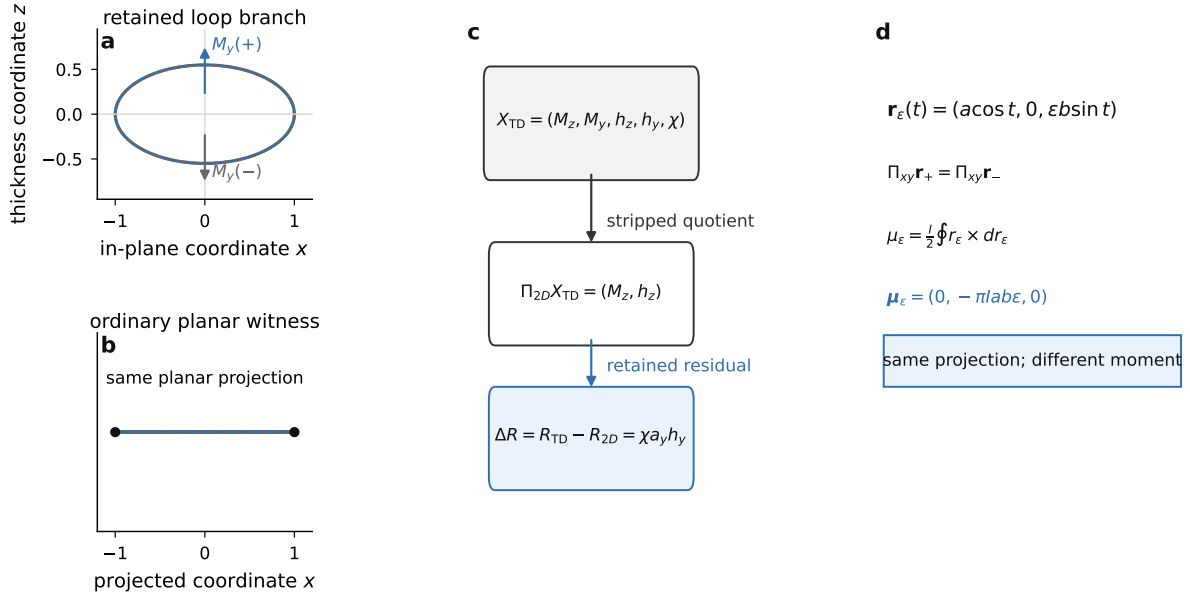


Figure 1: A through-thickness loop can share the same planar witness while retaining a different in-plane orbital moment. Panel a shows two opposite loop orientations in the thickness plane. Panel b shows that both orientations collapse to the same ordinary planar projection. Panel c shows the retained transdimensional state and the stripped two-dimensional quotient. Panel d gives the symbolic witness: the planar projection agrees, while the loop moment differs, leaving the residual contribution $\Delta R_{xy} = \chi a_y h_y$ in the transdimensional readout.

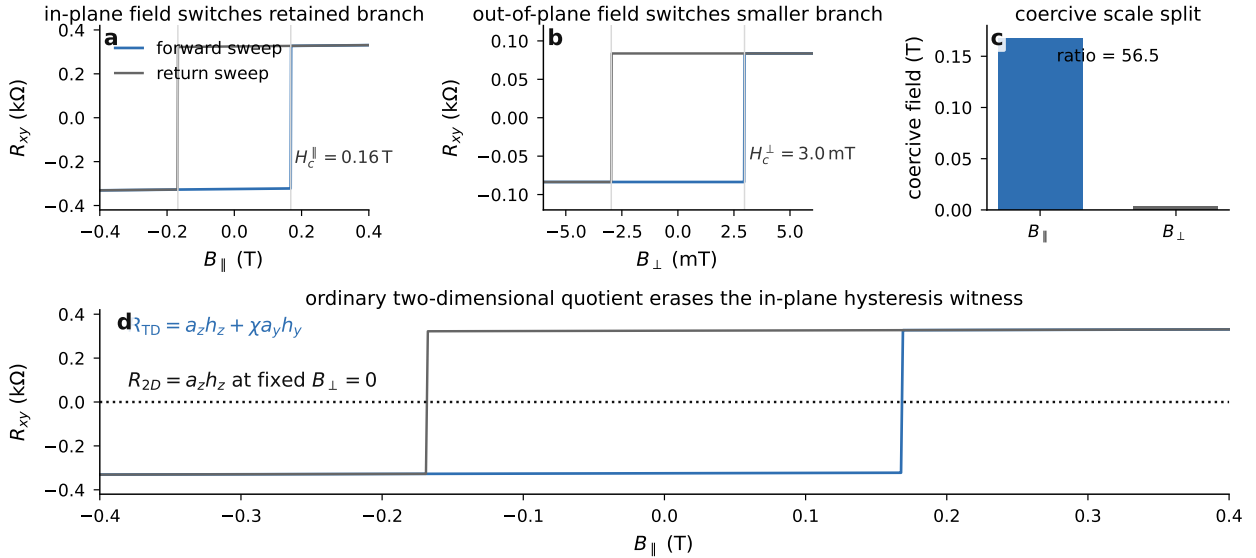


Figure 2: Retained branch memory reproduces concurrent in-plane and out-of-plane Hall hysteresis with separated coercive scales. Panel a shows the in-plane-field loop at $n = 1.40 \times 10^{12} \text{ cm}^{-2}$, $D = 0.90 \text{ V/nm}$, and $T = 0.05 \text{ K}$, with $H_{c,||} = 0.168 \text{ T}$. Panel b shows the out-of-plane-field loop with $H_{c,\perp} = 0.00297 \text{ T}$. Panel c shows the coercive-scale ratio, $H_{c,||}/H_{c,\perp} = 56.5$. Panel d compares the full transdimensional witness with the ordinary two-dimensional quotient at fixed $B_{\perp} = 0$; the stripped quotient removes the in-plane hysteresis witness.

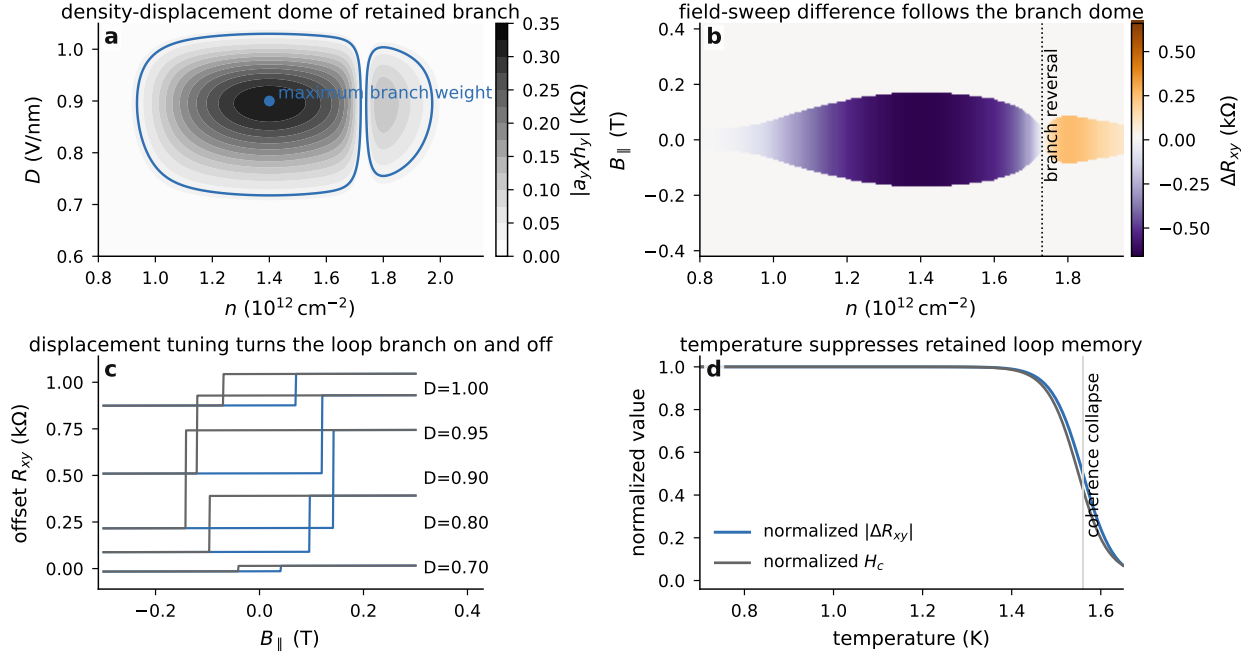


Figure 3: The retained in-plane loop branch appears only inside a finite density-displacement-temperature window. Panel a shows the branch-weight dome, with peak near $n = 1.40 \times 10^{12} \text{ cm}^{-2}$ and $D = 0.90 \text{ V/nm}$. Panel b shows the field-sweep difference map $\Delta R_{xy} = R_{xy}^{\uparrow} - R_{xy}^{\downarrow}$ at $D = 0.90 \text{ V/nm}$, including a high-density branch reversal near $n = 1.73 \times 10^{12} \text{ cm}^{-2}$. Panel c shows displacement-tuned parallel-field hysteresis loops, where the branch grows around $D = 0.90 \text{ V/nm}$ and fades near the edges of the window. Panel d shows temperature suppression of the normalized hysteresis amplitude and coercive field near 1.56 K.

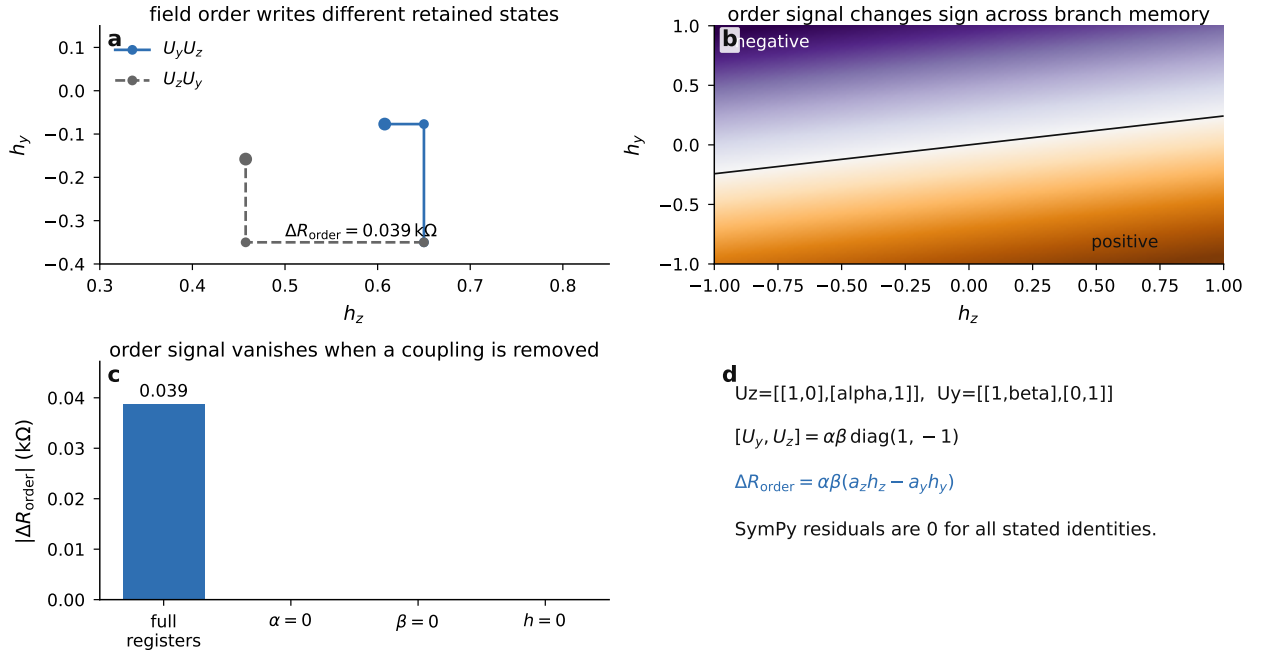


Figure 4: Field-training order produces a scalar Hall difference only when both retained branch-memory couplings are active. Panel a shows two field-order paths in the retained (h_z, h_y) register plane. Panel b shows the signed order signal ΔR_{order} across branch-memory states. Panel c shows that removing either coupling or the retained memory state reduces the order signal to zero. Panel d gives the finite-register commutator and the scalar order signal checked by symbolic simplification.

12 Experimental consequences of retained-branch transport

The retained-branch interpretation becomes useful only if it produces measurements that are difficult for a single scalar hysteresis model to absorb. The following consequences should be read as an experimental extension program: each one turns a Phase Calculus object into a directly measurable transport comparison.

12.1 Vector switching surface

The retained branch has a two-component switching condition. In the small-bias regime write the branch free-energy difference as

$$\Delta F = \Delta U(n, D, T, \eta) - 2M_z B_\perp - 2M_y B_\parallel, \quad \eta = \frac{d}{\ell_z}.$$

The switching surface is $\Delta F = 0$. Therefore the in-plane coercive field should shift under a fixed perpendicular bias as

$$B_\parallel^c(B_\perp) = B_\parallel^c(0) - \frac{M_z}{M_y} B_\perp, \quad M_y \neq 0.$$

The reciprocal measurement should hold on the same retained surface:

$$B_\perp^c(B_\parallel) = B_\perp^c(0) - \frac{M_y}{M_z} B_\parallel, \quad M_z \neq 0.$$

This is the cleanest direct test that the two hysteresis branches are components of one retained switching surface rather than unrelated scalar loops. The slope gives a transport estimate of the retained magnetization ratio:

$$\frac{\partial B_\parallel^c}{\partial B_\perp} = -\frac{M_z}{M_y}.$$

12.2 Field-order memory

Let $E_\parallel$ and $E_\perp$ denote branch updates induced by in-plane and perpendicular field training. A retained two-register branch state predicts

$$E_\parallel E_\perp X \neq E_\perp E_\parallel X$$

on reachable states inside the TDAHE regime. The measurable commutator is

$$\mathcal{C}_{\parallel\perp} = R_{xy}(E_\parallel E_\perp X_0) - R_{xy}(E_\perp E_\parallel X_0).$$

The prediction is

$$\mathcal{C}_{\parallel\perp} \neq 0$$

inside the TDAHE phase and approximately zero in neighboring ordinary metallic, quarter-metal, half-metal, or partially isospin-polarized phases. This measurement is stronger than comparing loop shapes because both protocols end at the same external field coordinates; the difference is carried by branch history.

12.3 Angular tomography of the retained loop branch

If the in-plane Hall branch is generated by a retained through-thickness orbital loop, rotating $B_\parallel$ within the plane should reveal the branch orientation. The residual

$$\Delta R_{xy}^{\text{TD}} = R_{xy}^{\text{TD}} - R_{xy}^{2D}$$

should be maximal when $B_{\parallel}$ couples to the retained loop axis and suppressed near the orthogonal direction. If multiple C_3 -related domains are present, the angular response should show domain-structured peaks rather than an isotropic in-plane signal. A convenient first fit is

$$\Delta R_{xy}^{\text{TD}}(\phi) = \Delta_0 + \Delta_3 \cos(3\phi + \phi_0) + \Delta_6 \cos(6\phi + \phi_1) + \dots,$$

with the branch-domain information carried by the residual rather than by the raw Hall trace alone.

12.4 Coherence-collapse scaling

The in-plane branch should collapse when vertical coherence is lost. Write

$$\eta = \frac{d}{\ell_z}.$$

The retained-branch amplitude should obey

$$A_{\parallel}(T, N, \mu) = A_0(n, D) \Phi\left(\frac{\ell_z(T, \mu)}{d_N}\right),$$

with

$$\Phi(x) \approx 0 \quad x \ll 1, \quad \Phi(x) > 0 \quad x \gtrsim 1.$$

Thus the collapse should organize by d/ℓ_z , not by temperature alone. The experimental paper already identifies a temperature-induced reduction in vertical mean free path as one possible route by which coherent out-of-plane orbital motion vanishes; the retained-branch prediction is that data from temperature, thickness, and disorder sweeps should collapse when expressed in this ratio.

12.5 Dual-memory minor-loop surface

A scalar hysteresis model uses one memory coordinate. The retained-branch model uses at least two:

$$R_{xy} = R_{xy}(h_z, h_y; n, D, T).$$

Partial $B_{\parallel}$ loops should change the subsequent $B_{\perp}$ loop, and partial $B_{\perp}$ loops should change the subsequent $B_{\parallel}$ loop. The observable should be a two-register memory surface, not a single coercive-field curve. A practical reconstruction is obtained by preparing a grid of partial-loop endpoints (h_z, h_y) , then measuring the local slope and curvature of R_{xy} across that grid.

12.6 First measurement package

A concise first request to an experimental group is the following. At a high-signal TDAHE point, for example near the reported large-signal region around $n \approx 1.4 \times 10^{12} \text{ cm}^{-2}$ and $D \approx 0.9 \text{ V/nm}$, measure R_{xy} while sweeping $B_{\parallel}$ under fixed perpendicular biases

$$B_{\perp} \in \{-8, -4, 0, +4, +8\} \text{ mT}.$$

Extract $B_{\parallel}^c(B_{\perp})$ and test the small-bias linear law above. Then reverse the experiment by sweeping $B_{\perp}$ under fixed $B_{\parallel}$ biases. Finally, compare endpoint-matched field-order protocols,

$$B_{\parallel} \rightarrow B_{\perp} \quad \text{and} \quad B_{\perp} \rightarrow B_{\parallel}.$$

The retained-branch prediction is a nonzero endpoint difference in R_{xy} inside the TDAHE phase and a near-zero difference in neighboring phases where the in-plane loop branch is absent.

12.7 Distinguishing outcomes

The retained-branch interpretation would be strongly disfavored by any of the following outcomes:

1. **No residual after proper branch separation.** If $R_{TD} - R_{2D}$ vanishes under all conditions where the reported in-plane hysteresis appears, then the retained residual model is not the right description.
2. **Endpoint-only memory.** If all field-order protocols with the same endpoints yield identical final retained states and identical minor-loop surfaces, then the monodromy-memory prediction is not observed.
3. **No coherence dependence.** If the in-plane branch survives unchanged when vertical coherence is destroyed, the through-thickness loop reading is strongly disfavored.
4. **No symmetry suppression.** If restoring the symmetry conditions that should forbid M_y leaves the residual unchanged, the branch-selector model is strongly disfavored.
5. **Scalar hysteresis sufficiency.** If a single scalar memory variable fits all major loops, minor loops, field-order protocols, and angular-domain switches with no residual, then the two-register branch-history model is unnecessary.

These distinguishing outcomes separate the structural claim from a loose interpretive analogy.

13 Verification and reproducibility support

The companion materials provide optional verification and reproduction support for the figure set and the new experimental-consequence identities. The deterministic figure data and executed notebook regenerate the four model-output figures. The symbolic checks return zero residual for the vector-switching surface identities, the reciprocal coercive-field shift, and the field-order commutator. A dependency-light Lean proof surface formalizes the finite retained-register claims: projection collision, branch-register noncommutation, and scalar-projection obstruction. The Lean material is included as a proof surface for Lean 4 verification; the local build environment used for this manuscript did not include a Lean executable, so the paper does not claim local Lean compilation.

14 Chronology and provenance

The mapping should not rest on chronology alone. It rests on theorem structure. Still, chronology matters for assessing whether the TDAHE reading was produced by adding new machinery after the fact.

The correct provenance claim is:

The TDAHE mapping uses Phase Calculus objects that are independently defined outside the TDAHE application: primitive roll, retained lifted state, operator core Q, B, L , CF000 saturation, quotient descent, and Monodromy-as-Memory. The mapping is legitimate if the TDAHE paper does not require adding new Phase Calculus machinery not already present in the source grammar.

A stronger historical-priority claim requires dated artifacts for each object. The public arXiv preprint for the TDAHE work is dated May 2025 [2]; the Nature version is dated 2026 [1]. Therefore a paper or source ledger should distinguish:

1. pre-arXiv Phase Calculus artifacts, if available;
2. pre-Nature Phase Calculus artifacts;
3. pre-mapping Phase Calculus artifacts;
4. later refinements that should not be used as evidence of prior prediction.

This draft makes the structural claim independent of personal discovery narrative. It uses definitions, theorems, and experimental consequences.

15 Discussion

15.1 Why the match is stronger than analogy

The strongest alignment is not verbal. It is algebraic.

The experiment reports a material regime where a lower-dimensional AHE description is incomplete because an additional orbital branch survives through finite thickness. Phase Calculus says that a visible projection can commute with part of a retained evolution while not being state-complete. Monodromy-as-Memory says that branch history is carried by a retained register and that stripped projection loses full-register information. The field-order matrices give a direct commutator that can be measured.

Thus the mapping has three independent layers:

retained loop theorem, quotient residual theorem, monodromy commutator theorem.

Each layer has an experimental correlate.

15.2 Why the ordinary AHE rule is not state-complete

The ordinary Hall rule is not false in its own regime. It is incomplete on the transdimensional branch. The issue is not that $\mathbf{E}_H \propto \mathbf{J} \times \mathbf{M}$ stops being useful. The issue is that the retained state contains an orbital loop component whose projection into the ordinary state variables is not faithful.

When $M_y \parallel J$, the naive cross-product expectation cannot account for a transverse Hall witness generated by the retained branch. The correct Phase Calculus reading is that the lower-dimensional law is a quotient law with a nonzero residual. The residual is not noise. It is the missing branch.

15.3 Why monodromy is the right memory language

Magnetic hysteresis is path memory. TDAHE has two field directions that can write different branch memories. Monodromy is the natural algebraic language for this because it treats loop/protocol traversal as an element of a group or semigroup acting on a retained state. The exact group model may be refined for real materials, where dissipation and domain dynamics may make the protocol monoid more appropriate than a group. The theorem-level burden remains: distinct protocol words can have distinct retained effects even when stripped projections identify them.

15.4 Relation to the experimental microscopic picture

The experimental paper's Hartree–Fock calculation identifies a symmetry-broken metallic state, a crescent-shaped Fermi surface, and pure orbital in-plane magnetization driven by electron–electron interactions [2, 1]. Phase Calculus does not replace those calculations. It classifies the resulting state: a retained orthogonal loop branch has become active, and the ordinary projection is incomplete.

That classification matters because it suggests what to measure next. Microscopic theory explains how a branch forms. Phase Calculus says what must happen when that branch is retained but projected through an incomplete quotient.

16 Conclusion

The transdimensional anomalous Hall effect is a concrete setting where a lower-dimensional visible law is incomplete because the physical state retains a loop branch outside the ordinary projector. Phase Calculus supplies a direct structural account of that incompleteness. CF000 describes the saturation pattern: when same-axis expression is exhausted and discharge is excluded, orthogonal re-articulation is forced. The retained-loop theorem shows how a through-thickness loop can have identical planar projection while carrying an in-plane orbital moment. The quotient-residual theorem identifies the missing Hall term. The Monodromy-as-Memory theorem turns hysteresis into a branch-history register and predicts field-order commutators.

The resulting extension program is experimentally sharp. Measure the residual, test field-order protocols, map the angular domain structure, collapse the branch by destroying vertical coherence, restore the symmetries that suppress M_y , and reconstruct the dual-memory minor-loop surface. If those patterns are observed, TDAHE is not merely an unusual Hall effect. It is a material witness of retained loop-generated projection beyond a saturated lower-dimensional quotient.

A Appendix A: compact theorem list

1. Visible recurrence is not retained identity:

$$V(\theta + 2\pi m) = V(\theta), \quad \kappa(\theta + 2\pi m) = \kappa(\theta) + m.$$

2. Through-thickness loop moment:

$$\gamma_\varepsilon(t) = (a \cos t, 0, \varepsilon b \sin t), \quad \mu_y = -\varepsilon \pi I a b.$$

3. Projection erasure:

$$P_0 \gamma_+ = P_0 \gamma_-, \quad \mu_y(\gamma_+) = -\mu_y(\gamma_-).$$

4. TDAHE residual:

$$\Delta_{\text{TD}} = R_{\text{TD}} - R_{2D} = \chi a_y h_y.$$

5. Monodromy composition:

$$E_{gg'} = E_{g'} \circ E_g.$$

6. Full-register faithfulness:

$$E_g = E_{g'} \iff g = g'.$$

7. Field-order matrix commutator:

$$U_y U_z - U_z U_y = \alpha \beta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

8. Scalar projection obstruction:

$$p_z(U_y(h_z, h_y)) = h_z + \beta h_y,$$

so no h_z -only quotient closes when $\beta \neq 0$.

B Appendix B: symbolic verification targets

The companion symbolic attack verifies:

1. the loop integral for μ ;
2. the equality of planar projections for γ_+ and γ_- ;
3. the residual identity $\Delta_{\text{TD}} = \chi a_y h_y$;
4. the field-order matrix commutator;
5. the Hall commutator under both sign conventions;
6. the scalar projection obstruction for p_z .

These checks do not prove the material experiment. They prove the algebraic identities used by the retained-state explanation.

C Appendix C: source ledger template

For publication, attach a dated source ledger with the following rows:

Phase Calculus object	Dated source	Role in this paper
Primitive roll	Phase Calculus formalisation / CF000	Generates visible witness from retained loop.
Lifted state $\hat{\Xi}$	Phase Calculus retained-state paper	State-complete carrier.
Operator core Q, B, L	Phase Calculus operator core	Same-host continuation, refinement, orthogonal lift.
CF000 saturation theorem	CF000	Forces orthogonal re-articulation after saturation.
Quotient criterion	Phase Calculus quotient paper	Tests whether visible law descends from retained evolution.
Monodromy-as-Memory	Monodromy paper	Stores branch-history words and predicts noncommuting protocols.

References

- [1] Qingxin Li, Hua Fan, Min Li, Yinghai Xu, Junwei Song, Anqi Wang, Kenji Watanabe, Takashi Taniguchi, Jing-Jing Chen, Zhenbing Tan, Jie Shen, Hua Jiang, James C. Hone, Cory R. Dean, Kostya S. Novoselov, Xin-Cheng Xie, Geliang Yu, Yue Zhao, Jianpeng Liu, and Lei Wang. Transdimensional anomalous Hall effect in rhombohedral thin graphite. *Nature*, 2026. DOI: 10.1038/s41586-026-10471-1.
- [2] Qingxin Li, Hua Fan, Min Li, Yinghai Xu, Junwei Song, Kenji Watanabe, Takashi Taniguchi, Hua Jiang, Xin-Cheng Xie, James C. Hone, Cory R. Dean, Yue Zhao, Jianpeng Liu, and Lei Wang. Transdimensional anomalous Hall effect in rhombohedral thin graphite. arXiv:2505.03891v1, 2025.
- [3] Naoto Nagaosa, Jairo Sinova, Shigeki Onoda, A. H. MacDonald, and N. P. Ong. Anomalous Hall effect. *Reviews of Modern Physics*, 82:1539–1592, 2010.
- [4] E. Hall. On the possibility of transverse currents in ferromagnets. *Philosophical Magazine*, 12:157–172, 1881.
- [5] Justin K. Lietz. Phase Calculus: complete formalisation of the primitive roll, exact lifted object, lifted-state operator core, selector-closed executable macro-calculus, universal lifted-state calculus, and Shadow Calculus. Neuroca, Inc. technical manuscript.
- [6] Justin K. Lietz. CF000 / primitive bifurcation and same-axis saturation: saturation theorem and orthogonal re-articulation. Neuroca, Inc. technical manuscript.
- [7] Justin K. Lietz. Monodromy-as-Memory: branch-history registers, full-register faithfulness, and quotient non-injectivity. Neuroca, Inc. technical manuscript.
- [8] Philip J. Morrison. Bracket formulation for irreversible classical fields. *Physics Letters A*, 100:423–427, 1984.
- [9] Miroslav Grmela and Hans Christian Ottinger. Dynamics and thermodynamics of complex fluids. I. Development of a general formalism. *Physical Review E*, 56:6620–6632, 1997.